

Node attributes and performance measures

Supply Chain Node Attributes		
Sr	Attribute/Feature	Description
1	Name	Name of the supply chain (SC) node
2	ID	Unique identifier for the node (alphanumeric)
3	Type	Type/category of the SC node (e.g., manufacturer, distributor)
4	Location	Geographic location (latitude, longitude) of the node
5	Inventory	Whether the node has inventory storage capability
6	Inventory Type	Type of inventory (e.g., finished goods, raw materials)
7	Inventory Product(s)	List of products that are stored in the inventory
8	Shelf Life	Time for which a perishable product stays usable at the node before it expires
9	Inventory Replenishment Policy	Replenishment policy (e.g., periodic, EOQ)
10	Inventory Capacity	Maximum inventory level that can be stored
11	Inventory Position	Inventory on hand plus inventory on order or in transit, minus backlog, tracked during the simulation and compared with the reorder level by reorder-point policies
12	Inventory Reorder Level	Threshold to trigger replenishment
13	Inventory Order-up-to Level	Target level to which inventory is raised when a replenishment order is placed
14	Order Quantity or Lot Size	Quantity ordered or produced per replenishment order or production run
15	Inventory Safety Stock	Buffer stock held to absorb demand and lead time variability
16	Inventory Reorder Period	Time interval for placing replenishment orders
17	Demand Forecast or Forecast Error	Forecast of the demand the node faces, with its error or bias, used to set orders and plans
18	Inventory Holding Cost(s)	Cost incurred per unit for holding inventory
19	Storage Cost	Cost of operating the storage facility, such as refrigerated storage in cold chains
20	Order Cost	Cost incurred per order placed to the supplier to replenish the inventory
21	Backorder Penalty	Penalty cost for delayed orders
22	Depreciation Cost	Decrease in product value over time due to wear, tear, or unfavorable market conditions
23	Establishment Cost	Cost of setting up the node (building the production plant or warehouse)
24	Selling Price	Price at which the node sells a unit of product, including markdowns
25	Probability of Node Failure	Probability of this node being unavailable due to power outage, equipment breakdown, or labour disruption
26	Disruption Duration	Time for which the node remains unavailable after a disruption
27	Recovery Duration	Time taken by the node to resume normal operation after a disruption
28	Disruption Cost	Cost incurred per disruption at this node, including the cost of recovery
29	Number of Backup Sources	Number of alternative suppliers or supply regions the node can draw on when its main source fails
30	Production Cost	Cost to produce a unit/batch of product (node-specific attribute: production nodes)

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31	Production Capacity	Maximum quantity the node can produce per unit of time (node-specific attribute: production nodes)
32	Vehicle Arrival Rate	Rate at which trucks, trains, or vessels arrive at the node (node-specific attribute: ports and terminals)
33	Processing or Service Time	Time to process, handle, load, or unload a unit or vehicle at the node (node-specific attribute: production and handling nodes)
	Supply Chain Node Performance Measures	
Sr	Performance Measure	Description
1	Throughput	Number of product units sold per unit of time (day/month/quarter)
2	Total sales/net sales	Total number of items sold over a given time period
3	Net profit	Net profit generated over a given time period (current month/quarter)
4	Average net profit	Average net profit per time unit (day/month/quarter)
5	Inventory levels	Inventory levels at regular time intervals (daily/monthly)
6	Total node cost	Total node cost (includes manufacturing, storing, moving, loading, unloading, and transporting)
7	Average node cost	Average node cost per unit time
8	Demand Placed	Number of orders (and product quantity) placed by this node
9	Demand Received	Number of orders (and product quantity) received at this node
10	Fulfillment Received	Number of orders (and product quantity) fulfilled for this node
11	Demand Fulfilled (Sales)	Number of orders fulfilled by this node
12	Customer Service Level	Percentage of demand fulfilled
13	Shortage	Number of orders not fulfilled by this node due to unavailability
14	Inventory Waste	Inventory waste due to product expiration (applicable for perishable products)
15	Disruption Waste	Quantity and monetary value of inventory destroyed by a disruption at this node
16	Inventory Carry Cost	Cost of storing each product item at the node
17	Inventory Spend Cost	Expenses for replenishing the inventory
18	Transportation Cost	Transportation cost paid for replenishment orders by this node
19	Revenue	Total income generated by this node
20	Production Level	Quantity of goods produced in a given time period
21	Supply Reliability	Probability of reliable supply at the node
22	Node Resilience	Node's ability to recover from disruption
23	Node Criticality	Refers to the importance of the node (e.g., a node that could cut the main flow of product if it is down)
24	Inventory Shortfall	Shortage amount (unmet demand quantity)
25	Backlog	Quantity of accepted orders not yet fulfilled, carried over and filled later rather than lost
26	Waiting/Turnaround Time	Time that vehicles, containers, or orders spend waiting or being handled at the node, such as dwell or turnaround time
27	Resource Utilization	Share of time that equipment, vehicles, or staff at the node are busy